

Adversarial Audit Report — Paper III

Linear-Error Clique Partitions of Split Graphs via Structured Triangle Packing

Version	v1.1.5
Date	2026-07-28
Auditor	Claude Opus 4.8 (Anthropic)

Cryptographic identifiers

Manuscript SHA-256

7aaf03083ddf7731dcb2b1e849cdfac97fb1697df1650c49a56e8431ce1bcb0b

Lean freeze ZIP SHA-256

060957e6b8d54779844dc6adf7cc7c3b8446fc17a87aa8d7a437e9d9d1001b78

GLOBAL VERDICT: PASS_WITH_OBSERVATIONS

Block verdict summary

The audit is organized into seven independent blocks (A–G). Each block was executed as an adversarial attack surface against the manuscript and its Lean formalization.

Block	Scope	Verdict
A	Claim Faithfulness	PASS
B	AX1/AX2 Literature Scope	PASS
C	Bulk/Sparse/Corridor Proof Attack	PASS
D	Counterexample & Boundary Search	PASS
E	Independent Computation	PASS
F	Audit the Internal Audit	PASS
G	Lean Verification	PASS_WITH_OBSERVATIONS

Key findings

(a) Core results sound (conditional)

Theorem 1.1 and Corollary 1.2 are mathematically sound, conditional on axioms AX1 and AX2. The proof assembly correctly discharges every intermediate lemma into the global conclusion.

(b) Axioms faithful, not overstated

AX1 and AX2 are faithful to the cited literature. AX2 is invoked in its *proven* form (density $0.9 + \epsilon$), not the merely *conjectured* $0.75n$ form — so no claim rests on an unproven strengthening.

(c) Corridor genuinely unconditional

The corridor argument is genuinely unconditional: its axiom footprints are closed, resting only on standard axioms with no residual dependence on AX1/AX2.

(d) One MINOR non-blocking observation (F-G05)

The Lean sparse node E_8 carries **AX1 + AX2**, not AX2-only as manuscript §11.6 states, because it reuses the bulk lemma E_4_3 on the $\alpha \in [1/12, 1/2]$ sub-range. This is a per-node attribution imprecision with **no mathematical impact** on any theorem or corollary.

Node axiom-footprint table

Per-node axiom footprints as verified in the Lean formalization. "Closed" denotes reliance on standard axioms only (no AX1/AX2).

Node	Axiom footprint
Theorem_1_1	AX1 + AX2

Node	Axiom footprint
Corollary_1_2	AX1 + AX2
E_4_3 (bulk)	AX1
E_8 (sparse)	AX1 + AX2
Prop_10_1_low	closed (standard axioms only)
Prop_10_1_mid	closed (standard axioms only)
packing corollaries	closed (standard axioms only)
Corollary_12_2_bound	closed (standard axioms only)
Phi_le_high_ratio	closed (standard axioms only)
cp_le_Phi	closed (standard axioms only)
nu3Star_le_tau3Star	closed (standard axioms only)